

Mean Value Theorems — Question Bank

Engineering Mathematics–I (CSE-1001)

BTech 1st Year, IIIT Bhopal

Kanchan Rajwar

kanchan.rajwar@iiitbhopal.ac.in

Instructions

This bank accompanies the note *The Mean Value Theorems*, the first note of Module 2.

How the parts are meant to be used:

1. **Part A (MCQ)** — rapid revision. One minute per question.
2. **Part B (True or False)** — every answer must carry a reason or a counterexample.
3. **Part C (Short answer)** — routine verifications, one theorem per question, exactly as they appear in an examination.
4. **Part D (Long answer)** — statements, proofs, and the longer application problems.
5. **Part E (Critical thinking)** — traps, counterexamples and faulty arguments to be diagnosed.

One habit worth forming. Before applying any of these theorems, write down the three hypotheses and check them one by one: continuity on the *closed* interval, differentiability on the *open* interval, and (for Rolle) equal endpoint values. A large share of the marks in this topic is for the checking, not for finding c .

A second habit. These are *existence* theorems. They promise that some c exists; they do not say where. In the verification questions of Part C you can solve for c because the functions are simple, but in a proof you may use only the fact that c lies strictly between the endpoints.

Answers and full worked solutions are issued as a separate document, *Mean Value Theorems — Answer Key*.

Part A: Multiple Choice

30 questions

Q1. Rolle's Theorem requires f to be continuous on

- (a) (a, b) (b) $(a, b]$ (c) $[a, b]$ (d) $\mathbb{R}$

Q2. The notation $(2, \infty)$ denotes

- (a) all real x with $x > 2$
(b) all real x with $x \geq 2$
(c) all real x with $2 < x \leq \infty$
(d) the two numbers 2 and ∞

Q3. In all three mean value theorems, the point c is guaranteed to lie in

- (a) $[a, b]$ (b) $\{a, b\}$ (c) $[a, b)$ (d) (a, b)

Q4. Differentiability is required only on the open interval (a, b) because

- (a) the theorems are easier to prove that way
(b) the conclusion concerns a point strictly inside the interval
(c) derivatives never exist at endpoints
(d) continuity already implies differentiability inside

Q5. Which function is continuous on $[-1, 1]$ but not differentiable on $(-1, 1)$?

- (a) $|x|$ (b) x^2 (c) x^3 (d) e^x

Q6. If f satisfies all three hypotheses of Rolle's Theorem, the conclusion is that for some $c \in (a, b)$

- (a) $f(c) = 0$ (b) $f''(c) = 0$ (c) $f'(c) = 0$ (d) $f(c) = f(a)$

Q7. For $f(x) = x^2 - 4x + 3$ on $[1, 3]$, the value of c given by Rolle's Theorem is

- (a) 1 (b) 2 (c) 3 (d) $\frac{5}{2}$

Q8. Rolle's Theorem fails for $f(x) = |x|$ on $[-1, 1]$ because

- (a) f is not continuous on $[-1, 1]$
(b) $f(-1) \neq f(1)$
(c) f is unbounded
(d) f is not differentiable at $x = 0$

Q9. The proof of Rolle's Theorem relies on which earlier result?

- (a) The Extreme Value Theorem
(b) The Intermediate Value Theorem
(c) Taylor's Theorem
(d) The Sandwich Theorem

Q10. Rolle's Theorem guarantees

- (a) exactly one such c
- (b) at most one such c
- (c) at least one such c
- (d) exactly two such c

Q11. For $f(x) = x^2 - 5x + 10$ on $[2, 3]$, the value of c in Rolle's Theorem is

- (a) 2
- (b) $\frac{5}{2}$
- (c) 3
- (d) $\frac{5}{4}$

Q12. Rolle's Theorem is the special case of Lagrange's Mean Value Theorem in which

- (a) $f(a) = f(b)$
- (b) f is a polynomial
- (c) f' is continuous
- (d) $b - a = 1$

Q13. Lagrange's Mean Value Theorem asserts the existence of $c \in (a, b)$ with

$$(a) f'(c) = 0 \quad (b) \frac{f(c)}{\frac{f(a) + f(b)}{2}} = (c) \frac{f'(c)}{f(a)} = f(b) - (d) \frac{f'(c)}{\frac{f(b) - f(a)}{b - a}} =$$

Q14. Geometrically, Lagrange's Mean Value Theorem says that somewhere inside the interval the tangent is

- (a) horizontal
- (b) vertical
- (c) parallel to the chord joining the endpoints
- (d) perpendicular to the chord joining the endpoints

Q15. For $f(x) = x^2$ on $[1, 3]$, the value of c in Lagrange's Mean Value Theorem is

- (a) $\frac{3}{2}$
- (b) 2
- (c) $\sqrt{5}$
- (d) $\frac{5}{2}$

Q16. For $f(x) = \ln x$ on $[1, e]$, the value of c is

- (a) 1
- (b) e
- (c) $\frac{e}{2}$
- (d) $e - 1$

Q17. The auxiliary function used to prove Lagrange's Mean Value Theorem from Rolle's is

- (a) $g(x) = f(x) - L(x)$, where L is the chord
- (b) $g(x) = f(x) + f'(x)$
- (c) $g(x) = f(x) L(x)$
- (d) $g(x) = f(x) - f(a)$

Q18. If $f'(x) = 0$ for all x in an interval I , then on I the function f is

- (a) zero
- (b) linear
- (c) constant
- (d) increasing

Q19. If $f'(x) > 0$ throughout an interval, then on that interval f is

- (a) positive
- (b) strictly increasing
- (c) concave up
- (d) bounded

- Q20.** If $|f'(x)| \leq M$ on (a, b) , then for $x_1, x_2 \in [a, b]$
- (a) $|f(x_2) - f(x_1)| \geq M|x_2 - x_1|$ (b) $|f(x_2) - f(x_1)| = M|x_2 - x_1|$ (c) $|f(x_2) - f(x_1)| \leq M|x_2 - x_1|$ (d) $|f(x_2) - f(x_1)| \leq M|x_2 - x_1|$
- Q21.** Cauchy's Mean Value Theorem requires, in addition to the usual hypotheses on f and g , that
- (a) g be a polynomial (b) $f'(x) \neq 0$ (c) $g'(x) \neq 0$ on (a, b) (d) $g(a) = g(b)$
- Q22.** Cauchy's Mean Value Theorem reduces to Lagrange's on taking
- (a) $g(x) = x$ (b) $g(x) = 1$ (c) $g(x) = f(x)$ (d) $g(x) = x^2$
- Q23.** For $f(x) = x^2$ and $g(x) = x^3$ on $[1, 2]$, Cauchy's Mean Value Theorem gives
- (a) $c = \frac{3}{2}$ (b) $c = \frac{14}{9}$ (c) $c = \frac{9}{7}$ (d) $c = \frac{7}{3}$
- Q24.** Which of the following is the *wrong* way to prove Cauchy's Mean Value Theorem?
- (a) Applying Rolle's Theorem to a suitable auxiliary function
 (b) Constructing $h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$
 (c) Checking that $h(a) = h(b)$ and invoking Rolle
 (d) Applying Lagrange's MVT separately to f and to g and dividing
- Q25.** In Cauchy's Mean Value Theorem the denominator $g(b) - g(a)$ cannot vanish because
- (a) g is assumed increasing
 (b) g is assumed positive
 (c) otherwise Rolle's Theorem would force g' to vanish somewhere
 (d) it is an extra hypothesis of the theorem
- Q26.** The inequality $|\sin x - \sin y| \leq |x - y|$ follows from the Mean Value Theorem because
- (a) $|\cos c| \leq 1$ (b) $\sin$ is bounded (c) $\sin$ is periodic (d) $\sin 0 = 0$
- Q27.** The inequality $|\tan x - \tan y| \geq |x - y|$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$ holds because
- (a) $\tan$ is odd (b) $\tan$ is unbounded (c) $\sec^2 c \leq 1$ (d) $\sec^2 c \geq 1$
- Q28.** Applying the Mean Value Theorem to $f(t) = e^t$ between 0 and x yields the inequality
- (a) $e^x \leq 1 + x$ (b) $e^x \geq 1 + x$ (c) $e^x \geq x$ (d) $e^x \leq ex$
- Q29.** Between any two consecutive real roots of a polynomial P there lies at least one real root of
- (a) P (b) P'' (c) P' (d) $1/P$
- Q30.** If $f'(x) > 0$ for every real x , then the equation $f(x) = 0$ has
- (a) at most one real root
 (b) at least two real roots
 (c) exactly one real root
 (d) no real root

Part B: True or False

10 questions — give a reason or a counterexample

- Q1.** If f is differentiable on (a, b) and continuous on $[a, b]$, then f' is continuous on (a, b) .
- Q2.** A function can satisfy the conclusion of Rolle's Theorem without satisfying its hypotheses.
- Q3.** If $f(a) = f(b)$ then $f'(c) = 0$ for some $c \in (a, b)$.
- Q4.** Rolle's Theorem guarantees exactly one point where the derivative vanishes.
- Q5.** If f is continuous on $[a, b]$ and $f(a) = f(b) = 0$, then f is identically zero on $[a, b]$.
- Q6.** The Mean Value Theorem tells us the value of c .
- Q7.** If $f'(x) = g'(x)$ for all x in an interval, then $f = g$ on that interval.
- Q8.** Lagrange's Mean Value Theorem can be applied to $f(x) = 1/x$ on $[-1, 1]$.
- Q9.** In Cauchy's Mean Value Theorem the same point c serves both functions.
- Q10.** If f is differentiable on $\mathbb{R}$ and f' is never zero, then f is one-to-one.

Part C: Short Answer

18 questions

- Q1.** State the three hypotheses of Rolle's Theorem and its conclusion.
- Q2.** Explain, in two sentences, why continuity is demanded on $[a, b]$ but differentiability only on (a, b) .
- Q3.** Give an example of a function that is continuous on $[0, 2]$, satisfies $f(0) = f(2)$, and for which Rolle's Theorem nevertheless fails. Say which hypothesis is violated.
- Q4.** Verify Rolle's Theorem for $f(x) = x^2 - 4x + 3$ on $[1, 3]$.
- Q5.** Verify Rolle's Theorem for $f(x) = x^2 - 5x + 10$ on $[2, 3]$.
- Q6.** Verify Rolle's Theorem for $f(x) = \frac{\sin x}{e^x}$ on $[0, \pi]$.
- Q7.** Verify Rolle's Theorem for $f(x) = (x - a)^m(x - b)^n$ on $[a, b]$, where m and n are positive integers.
- Q8.** Show that Rolle's Theorem does not apply to $f(x) = 4 - (6 - x)^{2/3}$ on $[5, 7]$, even though $f(5) = f(7)$.
- Q9.** State Lagrange's Mean Value Theorem and give its geometric meaning in one sentence.

- Q10.** Verify Lagrange's Mean Value Theorem for $f(x) = x^2$ on $[1, 3]$.
- Q11.** Verify Lagrange's Mean Value Theorem for $f(x) = \ln x$ on $[1, e]$.
- Q12.** Verify Lagrange's Mean Value Theorem for $f(x) = x^3 - 3x + 1$ on $[1, 3]$.
- Q13.** Verify Lagrange's Mean Value Theorem for $f(x) = \frac{x}{x-1}$ on $[2, 4]$.
- Q14.** Using the Mean Value Theorem, show that $|\sin x - \sin y| \leq |x - y|$ for all real x and y .
- Q15.** Using the Mean Value Theorem, show that $e^x \geq 1 + x$ for all real x .
- Q16.** Using the Mean Value Theorem, prove that for $0 < a < b$,
- $$\frac{b-a}{b} < \ln \frac{b}{a} < \frac{b-a}{a}.$$
- Q17.** Show that the equation $x^3 + x - 1 = 0$ has exactly one real root.
- Q18.** If $P(x) = (x-1)(x-3)(x+2)$, locate an interval containing a root of $P'(x)$ without computing P' , and then confirm your answer by computing it.

Part D: Long Answer

10 questions

- Q1.** State the Extreme Value Theorem. Explain precisely where it is used in the proof of Rolle's Theorem, and why the theorem would fail without it.
- Q2.** Prove Rolle's Theorem. In your proof, treat the constant case separately and justify carefully why the one-sided limits of the difference quotient at an interior maximum have opposite signs.
- Q3.** For each of the three hypotheses of Rolle's Theorem, give a function that satisfies the other two but not that one, and for which the conclusion fails. Explain each failure.
- Q4.** State and prove Lagrange's Mean Value Theorem, deriving it from Rolle's Theorem. Explain the geometric meaning of the auxiliary function you use.
- Q5.** Prove each of the following consequences of Lagrange's Mean Value Theorem:
- (i) if $f' \equiv 0$ on an interval then f is constant there;
 - (ii) if $f' > 0$ throughout an interval then f is strictly increasing there;
 - (iii) if $|f'| \leq M$ on (a, b) then $|f(x_2) - f(x_1)| \leq M|x_2 - x_1|$ on $[a, b]$.
- Q6.** Deduce from Lagrange's Mean Value Theorem that if two functions have the same derivative on an interval, they differ by a constant. Explain why this is exactly the statement that the indefinite integral is determined up to $+C$.

- Q7.** State and prove Cauchy's Mean Value Theorem. Verify that your auxiliary function satisfies all three hypotheses of Rolle's Theorem, and explain where the condition $g'(x) \neq 0$ is used.
- Q8.** Prove that $|\tan x - \tan y| \geq |x - y|$ for all $x, y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Compare the structure of this proof with that of $|\sin x - \sin y| \leq |x - y|$ and explain what decides the direction of the inequality in each case.
- Q9.** Prove that between any two consecutive real roots of a polynomial P there lies a real root of P' . Illustrate with $P(x) = (x - 1)(x - 3)(x + 2)$, computing the roots of P' explicitly and confirming that one lies in each gap.
- Q10.** Show that the equation

$$(x - 1)^3 + (x - 2)^3 + (x - 4)^3 + (x - 5)^3 = 0$$

has exactly one real root. Prove both that at least one root exists and that no second root can exist, and find the root exactly.

Part E: Critical Thinking

8 questions

- Q1.** A student writes: "*Rolle's Theorem needs f differentiable on $[a, b]$, so it cannot be applied to $f(x) = \sqrt{1 - x^2}$ on $[-1, 1]$.*" Is the conclusion right? Is the reasoning right? Answer both parts.
- Q2.** Consider $f(x) = 1 - |x - 1|$ on $[0, 2]$. Verify that $f(0) = f(2)$ and that f is continuous on $[0, 2]$, yet no $c \in (0, 2)$ has $f'(c) = 0$. Which hypothesis fails, and where exactly on the graph can you see the failure?
- Q3.** A student claims: "*If $f'(c) = 0$ for some $c \in (a, b)$, then $f(a) = f(b)$.*" Decide whether this is true, and either prove it or give a counterexample. Which theorem is being misread?
- Q4.** A student proves the Mean Value Theorem as follows:

"Since f is differentiable, $f(b) - f(a) = f'(c)(b - a)$ where c is the midpoint $\frac{a+b}{2}$."

Find a function and an interval for which this is false, and explain what the theorem actually claims about c .

- Q5.** A student argues: " *$f(x) = x^3$ has $f'(0) = 0$, so by Rolle's Theorem $x = 0$ is a maximum or a minimum.*" Diagnose the error.
- Q6.** A student presents the following proof of Cauchy's Mean Value Theorem:

"By the Mean Value Theorem, $f(b) - f(a) = f'(c)(b - a)$ and $g(b) - g(a) = g'(c)(b - a)$. Dividing the first by the second gives $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$."

Identify the error precisely. State what the argument does establish, and explain why the correct proof needs an auxiliary function.

Q7. A set of lecture notes states that between consecutive roots of $P(x) = (x - 1)(x - 3)(x + 2)$ the derivative has roots at approximately -0.76 and 2.426 . Verify or refute these two numbers by computing the roots of P' exactly. If one is wrong, say what the correct value is and whether the error affects the conclusion of the root-separation theorem.

Q8. A set of lecture notes asks for the number of real roots of

$$(x - 1)^3 + (x - 2)^3 + (x - 4)^3 + (x - 5)^3 = 0$$

and computes the derivative as $3[(x - 1)^2 + (x - 2)^2 + (x - 3)^2 + (x - 4)^2]$.

- (i) Identify the error.
 - (ii) Explain why the final conclusion happens to survive it.
 - (iii) Give a function and an equation for which the same slip *would* change the answer, and explain why.
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